CMSC 27100-3 Due: Friday, October 29, 2021

Problem Set 4

Instructor: Robert Rand

1 Induction II

Problem 1: We've stated in class that any finite set of natural numbers has a least element. Prove this using induction.

Problem 2: Let the *height* of a binary tree be the number of nodes on the longest path from the root to the leaves. Let the *size* of the tree be the total number of nodes. Prove that the size of a binary tree is always less than 2^{height}.

Problem 3: Two people are playing a game of hot potato. They can each hold the potato for 1, 2, or 3 seconds before passing it to their opponent. The potato starts with some n seconds on the clock (which both players know) and the player holding the potato when it hits 0 loses. (If a player throws it when it hits 0, the other player loses.) Under which conditions can the first player ensure victory? Prove it.

Problem 4:

(a) Use the principle of weak induction to prove the principle of strong induction.

(b) Use the principle of strong induction to prove the principle of weak induction.